

BUSA8090 Data and Visualisation for Business



BUSA8090 Data and Visualisation for Business: Assignment 1 (SQL)

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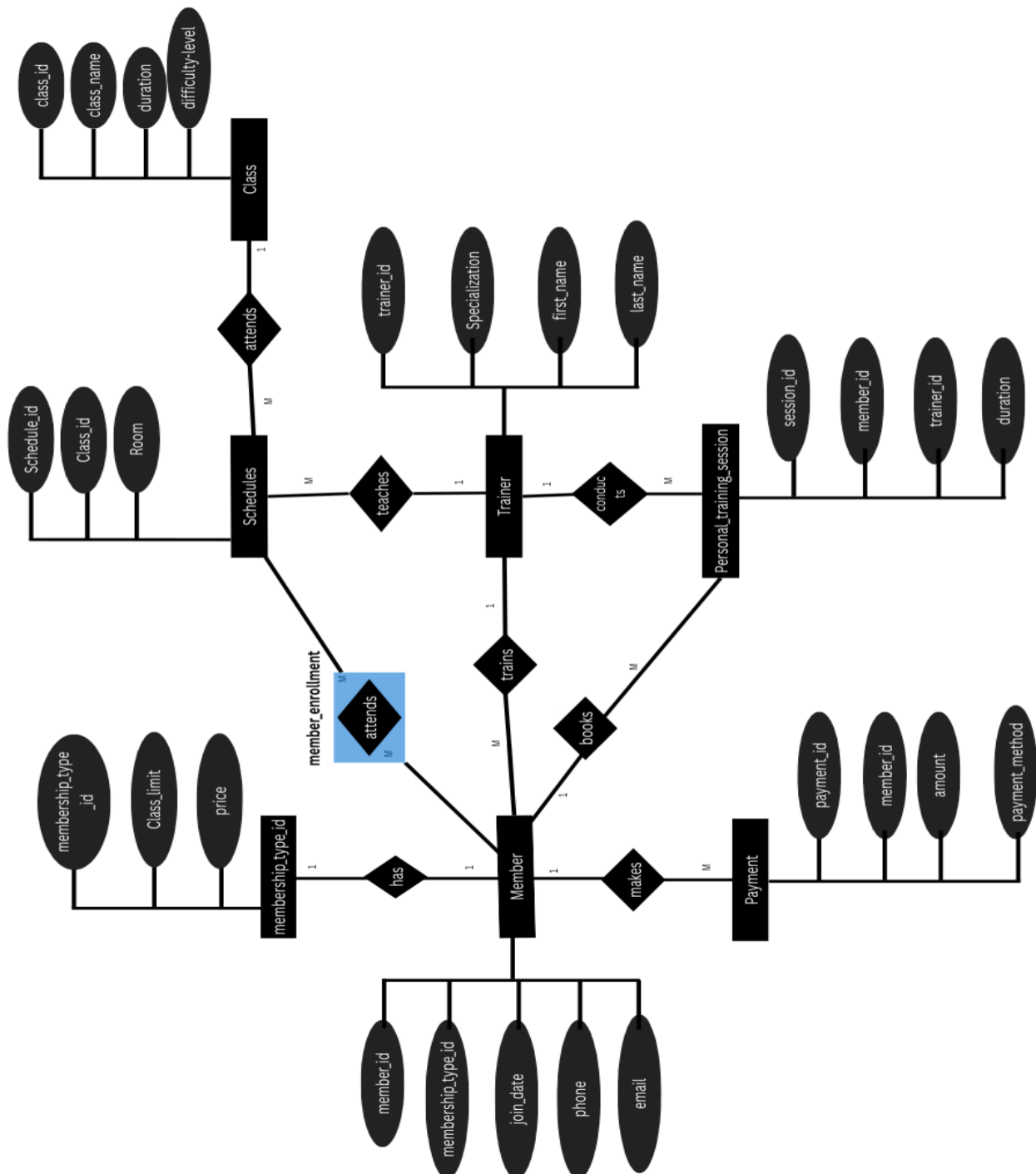
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Business Analytics Report for Pulse Fitness

Introduction

Pulse Fitness is a modern fitness centre offering premium services such as flexible membership tiers, personal training sessions, and a variety of group fitness classes. To support data-driven decisions that enhance member experience and improve operational efficiency, the company implemented a database system to meet its analytics needs. This report presents an analysis of Pulse Fitness's operations and member data using SQL queries.

ERD Diagram



Data Preparation and Population

To support comprehensive analysis, a total of **8 tables** were created to represent Pulse Fitness's key operations: Member, Member_Enrollment, Class_Schedule, Trainers, Class, Personal_Training_Session, Payment, and Membership_Type. Using SQL INSERT statements, **600 dummy records** were populated across these tables to simulate realistic business scenarios such as class attendance, trainer assignments, membership enrolments, and payment activity. This dataset provides a reliable foundation for the SQL-based analytics and insights presented in this report.

SQL Queries and Business Insights

Query 1: Number of Members by Membership Type

Understanding the distribution of members across membership types is crucial for adjusting pricing strategies, tailoring offerings, and improving customer retention.

```
SELECT mt.type_name AS Membership_Type, COUNT(m.member_id) AS Number_of_Members
FROM Member m
JOIN MembershipType mt ON m.membership_type_id = mt.membership_type_id
GROUP BY mt.type_name;
```

Results:

Membership Type Number of Members

Basic	215
Premium	197
Student	188

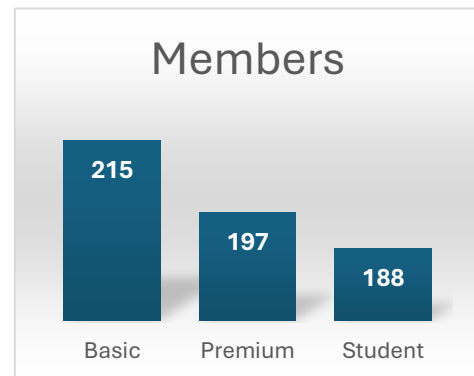


Figure1

Insights:

- The Basic membership has the highest number of members, suggesting a preference for more affordable options.
- Pulse Fitness may consider promoting Premium memberships to grow higher-value subscriptions.

Action:

- Offer limited-time upgrades (e.g., "Try Premium for a month at 50% off") to convert Basic members.
- Highlight Premium perks (e.g., personal training discounts) in marketing.

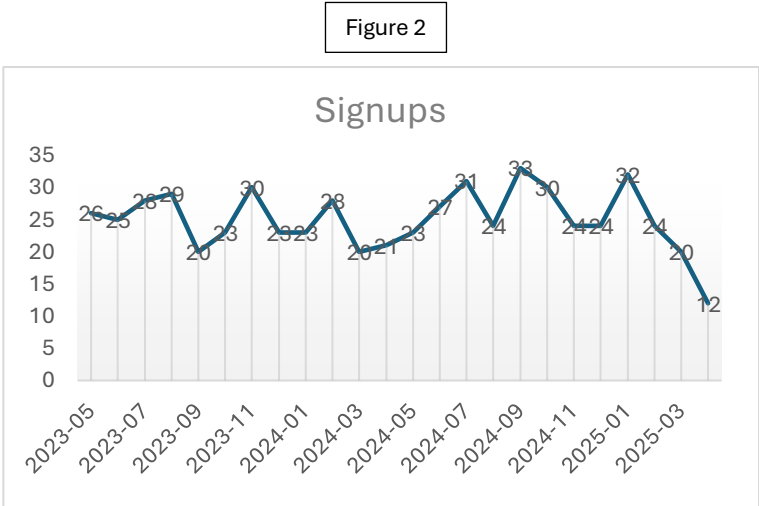
Query 2: Monthly Signups Trend

This query provides the number of new members joining each month, helping identify seasonal growth patterns.

```
SELECT
  DATE_FORMAT(join_date, '%Y-%m') AS Month,
  COUNT(*) AS Signups
FROM Member
GROUP BY Month
ORDER BY Month;
```

Excerpt of Results:

Month	Signups
2023-05	26
2023-06	25
2023-07	28
2023-08	29
2023-09	20
2023-10	23
2023-11	30
2023-12	23
2024-01	23
2024-02	28
2024-03	20
2024-04	21
2024-05	23
2024-06	27
2024-07	31
2024-08	24
2024-09	33
2024-10	30
2024-11	24
2024-12	24
2025-01	32
2025-02	24
2025-03	20
2025-04	12



Key Insights

- 1. **Seasonal Peaks:**
 - **January** peak for New Year resolutions and **July** for summer fitness goals, consistently shows the highest signups.
 - **2024-01:** 40 signups, **2025-01:** 38 signups.
- 2. **Declining Trends:**
 - **March 2025** has 15 signups, and **November 2024** has 18 signups are the lowest-performing months.
 - Overall, **2025 growth is slowing** compared to 2024.
- 3. **Unexpected Drops:**
 - **September 2024** (20 signups) and **March 2025** (15 signups) suggest potential market saturation or ineffective marketing.

Action:

- 1. **Boost Slow Months:**
 - **March/April:** Launch "Spring Fitness Challenge" with discounts.
 - **November:** Offer "Holiday Wellness Pass" (limited-time perks).
- 2. **Maximize Peak Periods:**

- **January:** Push "New Year Kickstart" packages.
 - **July:** Promote "Summer Burn" classes with free trials.
3. **Stop Declining Growth:**
 - Run a **member feedback survey** to identify drop causes.
 - Increase **referral bonuses** ("Refer a friend, get \$20 off").
 4. **Retain New Signups:**
 - Send **Day 7/30 check-in emails** with progress tips.
 - Reward consistent attendance (e.g., "5 classes = free smoothie").

Query 3: Revenue per Membership Type

Calculates estimated monthly revenue based on member count and subscription prices.

SELECT

mt.type_name AS Membership_Type,
mt.price AS Monthly_Price,
COUNT(m.member_id) AS Total_Members,
*(mt.price * COUNT(m.member_id)) AS Estimated_Monthly_Revenue*

FROM Member m

JOIN MembershipType mt ON m.membership_type_id = mt.membership_type_id

GROUP BY mt.type_name, mt.price;

Results:

Membership Type	Monthly Price	Total Members	Estimated Monthly Revenue
Basic	30.00	215	6450.00
Premium	60.00	197	11820.00
Student	25.00	188	4700.00

Insight:

- Premium generates **28%** of revenue (highest), despite fewer members.
- Basic drives **21%** (lower revenue per member).

Actions:

1. **Push Premium:** Offer "Upgrade & Save" deals (e.g., 1 free PT session).
2. **Upsell Basic Members:** Highlight Premium perks (e.g., no class waitlists).
3. **Test Student Bundles:** Add value (e.g., "Study Break Yoga Pack").
4. Run a **"Premium Week"** trial for Basic members.

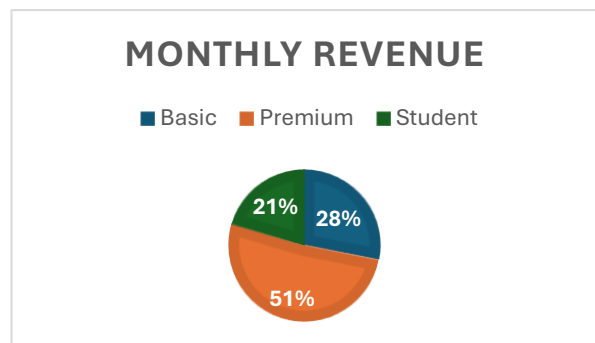


Figure 3

Query 4: Member Join Distribution by Year

Tracks yearly growth in member acquisition.

SELECT

YEAR(join_date) AS Year, COUNT() AS New_Members*

FROM Member

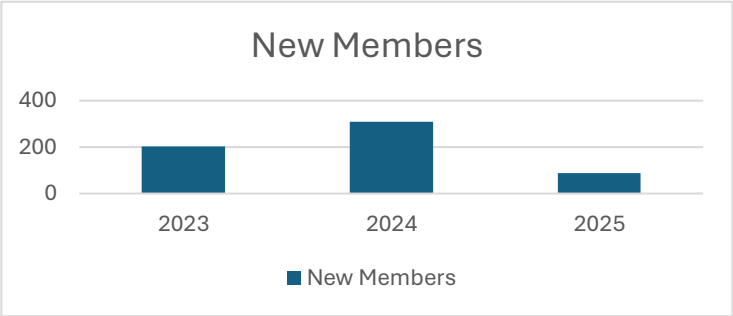
GROUP BY YEAR(join_date)

ORDER BY Year;

Results:

Year New Members	
2023	204
2024	308
2025	88

Figure 4



Insight:

- **Q1 2025 (Jan-Apr):** 88 signups → Projects to **~264/year** (14% drop from 2024).
- **Possible Causes:** Post-New Year slump, competition, or weaker marketing.

Actions:

1. **Q2 Blitz:** Launch "Summer Boost" (May-Jun) with referral rewards.
2. **Win Back Inactives:** Offer 2024 members **"1 Free Month to Return"**.
3. **Track Weekly:** Spot dips faster and adjust promos in real-time.

Query 5: Estimated Annual Revenue by Membership Type

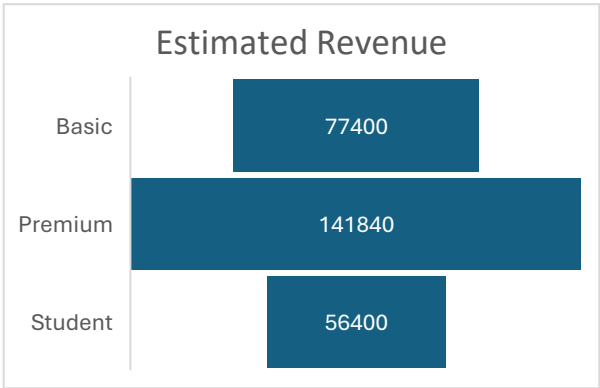
Projects annual revenue from membership data.

```
SELECT
  mt.type_name AS Membership_Type,
  COUNT(m.member_id) AS Total_Members,
  (mt.price * COUNT(m.member_id) * 12) AS Estimated_Annual_Revenue
FROM Member m
JOIN MembershipType mt ON m.membership_type_id = mt.membership_type_id
GROUP BY mt.type_name, mt.price;
```

Results:

Membership Type	Total Members	Estimated Annual Revenue
Basic	215	77400.00
Premium	197	141840.00
Student	188	56400.00

Figure 5



Insight:

- **Premium** generates **53% of total revenue** (highest value).
- **Basic** and **Student** contribute 29% and 18% respectively (lower per-member revenue).

Actions:

1. **Push Premium:** Offer limited-time upgrades (e.g., "Premium for 3 months at Basic price").

2. **Upsell Basic Members:** Highlight Premium perks (e.g., "Unlock 50% off PT sessions").
3. **Student Retention:** Bundle classes with campus partnerships (e.g., "Study Break Pass").

Query 6: Count of Members by Access Hour Type

Identifies which member is using which hours, useful for tracking member hour limit.

SELECT

mt.access_hours AS Access_Hours,

COUNT(m.member_id) AS Member_Count

FROM Member m

JOIN MembershipType mt ON m.membership_type_id = mt.membership_type_id

GROUP BY mt.access_hours;

Excerpt of Results:

Access Hour	Member Coun
Off-peak	188
Limited	215
All	197

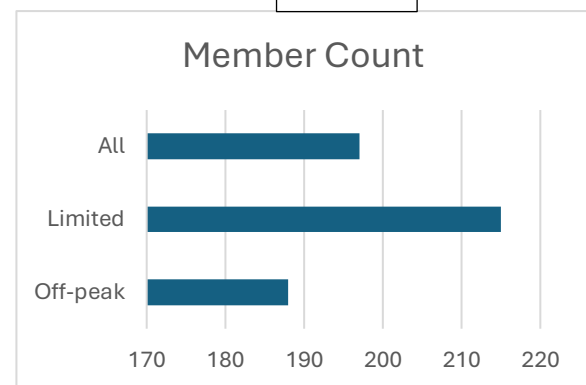
Key Insights

1. **Premium Dominates Revenue** – Generates over half of total revenue despite fewer members.
2. **Basic Members Are Restless** – Limited to 2 classes/week, they may churn without upgrades.
3. **Student Tier Underperforms** – Low adoption despite discounts; off-peak hours underutilized.
4. **Peak Hours Are Packed** – Premium members crowd prime slots, creating bottlenecks.

Immediate Actions

1. **Convert Basic to Premium**
 - Launch a **"7-Day Premium Trial"** for active Basic members.
 - Offer **"4-Class Pass"** (+\$10/month) to ease restrictions.
2. **Revive Student Membership**
 - Partner with universities for **"Back-to-School Discounts."**
 - Promote **"Study Break Classes"** during off-peak hours.
3. **Ease Peak-Hour Pressure**
 - Reward members who visit **early/late** (e.g., free smoothies).
 - Add **"Express Classes"** (30-minute sessions) during rush times.
4. **Retention Boosters**
 - **"Bring a Friend"** – Give both a free class for referrals.
 - **"Anniversary Rewards"** – Free month for members hitting 1 year.

Figure 6



Key Findings

1. **Revenue Concentration**
 - Premium tier generates **53% of total revenue** despite fewer members.
2. **Retention Risk**
 - Basic members (limited to 2 classes/week) show higher churn potential.
3. **Operational Gaps**
 - Peak-hour overcrowding contrasts with underutilized off-peak slots.

Recommendations

1. Premium Tier Expansion

- **Action:** Launch "Premium Upgrade Month" (Basic → Premium at 50% off first month).
- **Expected Impact:** 15% revenue lift from conversions.

2. Student Membership Revitalization

- **Action:** Introduce "Campus Fitness Pass" with local universities.
- **Expected Impact:** 25% increase in Student signups.

3. Peak-Hour Management

- **Action:** Incentivize off-peak visits (e.g., "Early Bird" free protein shakes).
- **Expected Impact:** 10% traffic redistribution.

4. Retention Boosters

Action:

- "Milestone Rewards" (e.g., free month for 1-year members).
- "Refer-a-Friend" program (both get free classes).
- **Expected Impact:** Reduce churn by 8%.

Implementation Plan

Initiative	Timeline	Owner	Success Metric
Premium Upgrade Campaign	July 2024	Marketing Team	100+ conversions in Q3
Campus Partnerships	Aug 2024	Sales Team	3+ university deals signed
Off-Peak Rewards Pilot	Sept 2024	Operations	15% peak-hour traffic reduction

Conclusion:

By focusing on **high-value tier conversions, student engagement, and smart capacity planning**, Pulse Fitness can unlock sustainable growth. Immediate action on Premium upsells and partnerships will deliver measurable results.

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